

1. Workload

2. ISA & ABI

3. Compute Org

4. Memory & NoC

5. Power Intent

6. Compiler/RT

7. Physical Node

8. Verification



CHAPTER STACK SCOPE — ACTIVE LAYERS 1-5 (WORKLOAD TO RTL)

Architecture Representations & World Models

1. Intent Specs

- Structured Prompt JSON
- Architecture Constraints
- Workload Encodings
- System Obligations

2. Code & Graphs

- Python / C++ Sim Models
- Chisel / SystemVerilog RTL
- Dataflow IR & AST Graphs
- Architecture Knowledge Base

3. World Models

- Surrogate Latency Predictor
- Power / Area Surrogate Model
- Placement Heatmaps
- Uncertainty Interval Guard

4. Validated Reps

- Schema Validation Check
- Structural Completeness
- Handoff Artifact Card
- Contract Compliance

Representation Refinement & Model Calibration Loop